OPEN AGENT HACKATHON 2026 · TRACK 02 · THE AGENT THAT CAN EXPLAIN WHY

# x402 Adoption Investigator

A multi-agent system that answers "how real is x402 adoption?" and proves every number it says: each claim links to a URL with the hash of the bytes fetched, a Base transaction, or a line of its own log.

x402 = pay-per-call over HTTP 402 Payment Required, settled in USDC on Base.

Team Automata · Martin Pisacreta (solo, Argentina) · Zetaris · Meterless · built with Claude Code and Cursor Agent

THE USER AND THE PROBLEM

# One store, five sources, five answers

User: the operator of an autonomous agent that sells or buys over x402 and must decide where the real demand is. Asked how many paid endpoints one store has, three sources say 17, two say 19 and 36, and none of them says why.

| SOURCE | WHAT IT MEASURES | ENDPOINTS FOR TIENDA.TEXTOKIT.STORE |
|---|---|---|
| CDP Bazaar search | listed resources, capped at 20 results | **17** |
| Store openapi.json | declared paid operations | **17** |
| Agent402 | paid tools (of 34 tools) | **17** |
| vet402 | resolved endpoints | **19** |
| x402-trust | keyword search matches | **36** |

Activity disagrees too: Bazaar 30-day counters sum to 60 calls, the seller's receipts list 77 settled sales, and 56 of them were paid by the seller itself.

WHY THIS NEEDS AGENTS

# A script would pick one number. This system explains all of them.

### Live, conflicting sources

Bazaar, the seller's receipts, Base RPC, x402-trust, Agent402 and vet402 each measure a different thing on a different window.

### On-chain verification

Settlements are checked on Base by decoding the USDC Transfer log, never trusting tx.from, which is the facilitator.

### Disagreement as output

When sources differ, the answer carries a Conflict object with each value, its measurement basis and why they differ.

### Failure is normal

Sources go down mid-run. They become explicit "unavailable" entries with a logged recovery, never silent zeros.

WHAT GOES IN, WHAT COMES OUT

# A question in, a proof-carrying answer out

Input: one JSON question

```
{"run_id": "ex01-seller-audit",
 "input": {"question": "How real is the adoption of
   tienda.textokit.store? Who pays, how much,
   do they come back, and why do registries
   disagree about it?"}}
```

Output: one JSON envelope and a Markdown report

- **verdict** chosen by rules, never by a model
- **claims[]** each with evidence: URL + sha256, Base tx, or log line
- **conflicts[]** with per-source values and the explanation
- **rejected_claims[]** with the critique trail
- **unavailable[]** instead of zeros

ARCHITECTURE

# Seven roles, bounded loops, every handoff logged

**Planner**
picks the playbook, sources and queries

**Collector**
fetches every source and hashes the bytes

**Keeper**
folds evidence into the World Model

**Verifier**
decodes settlements on Base

**Analyst**
computes each metric in SQL and in Python

**Writer**
drafts an answer that cites claims

**Critic**
rules C1 to C7: rejects or sends back

## Computed twice

Every metric has a committed SQL query and a pure Python reducer over the same exports. A number survives only if both exist and agree (rule C2); a mismatch stays as an open Conflict.

## Bounded and logged

Deadline, fetch, SQL, model-call and critic-round budgets. Every handoff, fetch, sql, claim, conflict, critique, retry and recovery line goes to logs/<run_id>/run.jsonl.

> Planner, Writer and the Critic's second opinion can call an OpenAI-compatible model. No model key was used in any committed run, so they ran on rules and templates.

A REAL RUN, CONDENSED

# What a judge sees in run.jsonl

```
guard     network disabled in sample mode (socket.connect patched)
handoff   planner → collector   plan (10 items, sha256 abce43a7…)
handoff   collector → keeper    observations (9, sha256 debfb264…)
recovery  collector  fixture_miss on a Base receipt: settlement marked unverifiable
handoff   verifier → keeper     chain_facts (1)
handoff   keeper → analyst      query_ids (13)
conflict  k01 endpoint_count_for_host   five sources, five measurement bases
conflict  k02 calls_30d_for_host        Bazaar 30-day counter vs seller receipts
handoff   writer → critic       answer
critique  round 1  C1…C7 accept 19 claims · narrative pass: 62 numbers mapped, 0 unsourced
handoff   keeper → hmem         claims (19) · memory feedback written
```

Condensed from the 108 JSON lines of example 01: 11 handoffs, 17 fetches, 13 SQL queries, 8 critiques, 5 recoveries, 2 conflicts.

CRITIQUE AND RECOVERY

# The Critic says no, and the answer says why

```
claim      c07 verified_tx = 0   sql 0 · reducer 0
critique   round 1  C1 reject c07: no decoded
           settlement of 0xa1d1…1978 (1 sampled:
           unverifiable, fixture_miss)
critique   round 1  B revise: sentences cite c07
handoff    critic → writer   issues (5)
recovery   writer  narrative_trimmed: c07 left
           out of the answer and named
critique   round 2  C1…C7 accept 6 · B accept
result     partial · 6 accepted · 1 rejected
output     verdict insufficient_evidence (by rules)
```

### Why the zero is rejected

SQL and reducer agree on 0, but nothing was decoded for this wallet. Zero verified would be a number about missing data, so the Critic reports it as unknown.

### Not staged

Same question as example 03, another external payer of the same store. Live, its settlement is verified at block 52,914,075.

Condensed from the 83 JSON lines of example 04: 13 handoffs, 20 critiques, 6 recoveries, 2 critic rounds.

METERLESS, STRUCTURAL

# World Model as shared state, H-MEM as policy

## World Model

- Entities: sellers, endpoints, wallets, facilitators, registries, with bitemporal relationships
- Provenance on every row; a claim without a URL, tx or log line is refused by the engine
- Idempotent ingest, proved in every output: the second pass appends 0 events
- 8 of 8 conformance checks

## H-MEM

- Seven pitfall memories ("never sum unique payers across endpoints", "receipts are self-reported") add caveats that Critic rule C3 checks
- Injected into Planner and Writer prompts when a model is mounted, logged by hash; tested offline
- Ablation committed: one memory less drops a caveat and fifteen citations, not the verdict
- 13 of 13 upstream vectors

ZETARIS, THE LIVE FEDERATION LAYER

# Live on Lightning SQL, offline on SQLite, same rows

- Each live run registers its exports as Lightning REST tables (receipts, Bazaar, openapi, Agent402, vet402, x402-trust, settlements) and flattens them with SchemaStore views
- The live seller audit computed all 19 claims on Zetaris 2.4.3.1
- Offline, the same SQL runs on SQLite; a test proves equal rows against the 19 result sets Zetaris recorded

### Logged per query

Executor, SQL sha256, rows, time and tables with source URLs in sql.jsonl; every REST call in zetaris.jsonl. --require-zetaris turns any fallback into an error.

### Not yet on Zetaris

3 of the 22 queries (q20 to q22) were added after the live session and have run on SQLite only.

RESULTS ON REAL DATA

# Four questions, four honest answers

### Seller audit · self_settled_mostly

77 sales, 56 paid by the seller's own wallets; 21 external sales by 4 payers, 0.044 USDC in total.

### Catalog · concentrated

28,462 Bazaar resources on 2,680 hosts; the top host holds 29.5% of 824,796 calls in 30 days.

### Payer dossier · adoption_thin

One purchase of 0.001 USDC, its settlement decoded and verified on Base against the receipts.

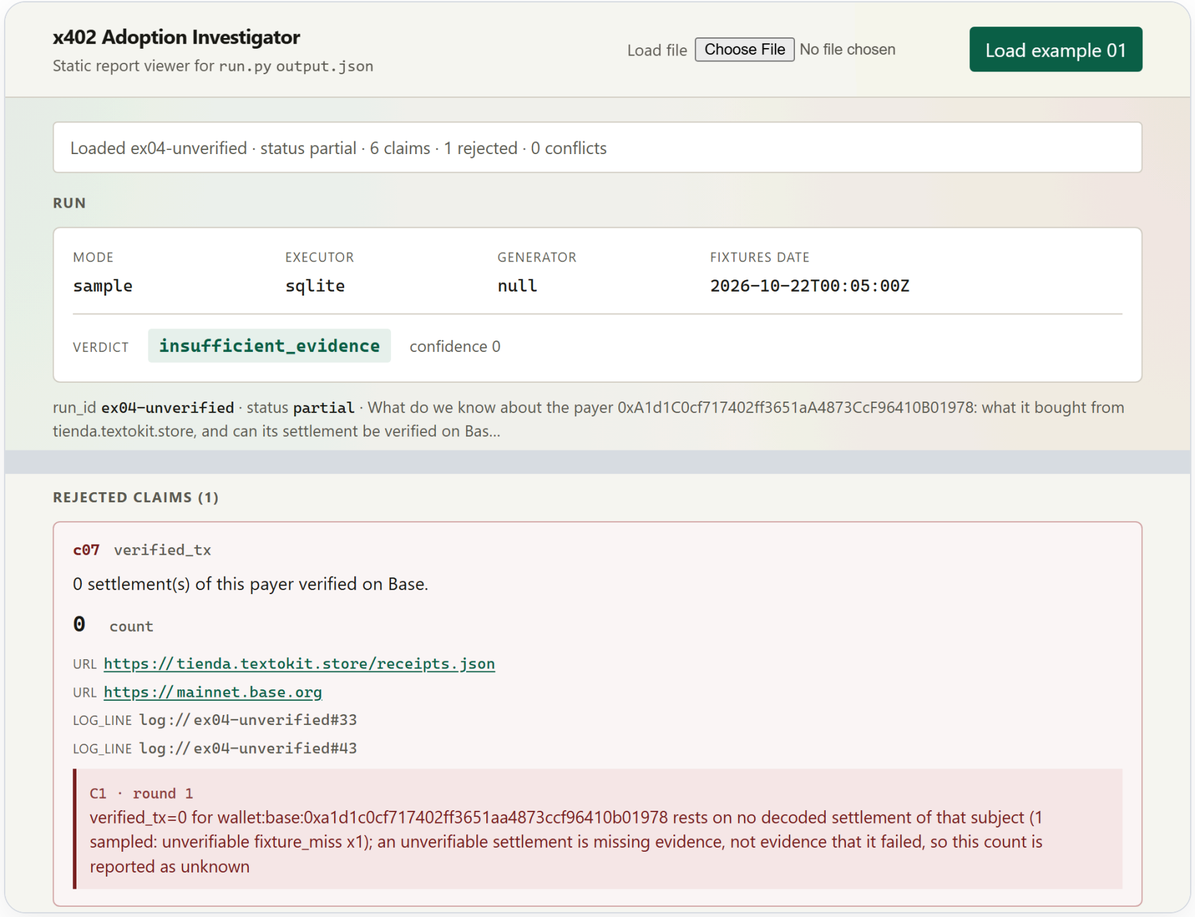
### Unverifiable payer · insufficient_evidence

One purchase of 0.001 USDC; "0 verified" rejected by the Critic, balance and nonce unavailable.

The honest finding for our own store: most of its sales were paid by its own wallets. The system says so, with the proof.

SEE IT

# The report viewer




## Any output.json

Run header with the mounts, accepted claims with their evidence links, rejected claims with the Critic's reason, and conflicts with each source's value and explanation.

Static HTML, no build step, no CDN. Shown: example 04 in sample mode, two crops of one page (header, rejected claims).

STACK AND HOW TO TRY IT

# One command, offline, from a clean clone

```
git clone https://github.com/martinPisacreta/x402-adoption-investigator
cd x402-adoption-investigator
python run.py --sample --input examples/ex4.json
```

**Python 3.11, standard library** | **Zetaris Lightning 2.4.3.1** | **Meterless World Model + H-MEM** | **Base JSON-RPC** | **Docker** | **225 tests**

**Claude Code: main coding agent** | **Cursor Agent: viewer, README pitch draft** | **NVIDIA: not used**

No model ran in the committed runs: the model path is implemented and tested offline with an OpenAI-compatible double.

Pre-existing, declared: our own Automata agent and its x402 store tienda.textokit.store (data source only, none of its code here), the Meterless conformance vectors (copied with attribution), the Zetaris platform and the Python standard library.